PRODUCT REQUIREMENTS DOCUMENT

Onigiroku — Japanese Learning, Practice, Assessment & Progress Intelligence SaaS

Including System Requirements Specification (SRS)

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Document purpose — This document defines the product scope, user journeys, functional and non-functional requirements, learning analytics model, architecture direction, security baseline, data model, release roadmap, and acceptance criteria needed to build Onigiroku from inception to production.

Document Control

Item	Definition
Product	Onigiroku
Document type	Product Requirements Document (PRD) with embedded System Requirements Specification (SRS)
Target readers	Product owner, system analyst, UI/UX designer, software engineers, QA engineers, DevOps/SRE, content team, finance and operations
Baseline scope	Responsive web SaaS for Japanese-language learning, exam practice, quizzes, learning notes, progress tracking, subscriptions and content administration
Change management	Material scope changes require version increment, impact analysis and approval before implementation
Normative language	“Shall” indicates mandatory requirement; “should” indicates recommended requirement; “may” indicates optional capability

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Executive Decisions

Decision	Recommended baseline
Access model	Three access states, but only two authenticated authorization roles: ADMIN and USER. “General user” is an anonymous/public access state, not a persisted role.
Authentication	Use Better Auth as the single source of truth for credentials, sessions, email verification and Google OAuth. Do not run Firebase Authentication in parallel unless a separate Firebase-dependent requirement emerges.
Subscription	One-day trial starts after verified registration. Paid entitlement is monthly. Non-tokenized QRIS/VA/e-wallet payments should be treated as manual renewal; recurring billing should only be enabled for supported tokenized payment methods.

Decision	Recommended baseline
Payment currency	Midtrans charge requests are sent in IDR. International visitors may see an indicative local-currency display, but checkout and accounting remain IDR unless a second international payment processor is added.
Progress retention	Never delete historical learning evidence merely because a learner becomes inactive. Maintain immutable history; calculate an "active proficiency profile" with recency decay and create comparison snapshots.
Analytics integrity	Separate assisted attempts from independent attempts. Answer reveals, hints, pauses, retries and time-on-task shall be recorded as first-class learning events.
Content governance	Questions and learning materials use draft → review → scheduled/published → archived lifecycle with versioning and audit trail.
Architecture	Modular monolith first, with clearly separated domains and background workers. Split into services only after measurable scaling or organizational need.

1. Product Overview

1.1 Product Vision

Onigiroku is a trusted, measurable and accessible self-learning platform that helps learners prepare for Japanese-language examinations, strengthen vocabulary and scripts, study structured learning notes, and understand precisely how their capability changes over time. The product differentiates itself by combining high-quality content with evidence-based learning analytics rather than presenting only scores.

1.2 Problem Statement

- Learners often practice from disconnected sources and cannot see a unified record of strengths, weaknesses and learning consistency.
- Conventional quiz systems overstate ability because they do not distinguish independent answers from answers completed after hints or answer reveals.
- Exam preparation requires both realistic timed simulation and lower-stakes formative practice with immediate explanations.
- Learning materials, practice history and exam outcomes are rarely connected at topic, competency and JLPT level.
- Content administrators need a governed workflow to continuously add and revise questions, explanations, media and learning notes without code changes.

1.3 Product Goals

ID	Goal	Success signal
G-01	Provide credible exam preparation and formative quizzes across Japanese-language competency areas.	Users can complete structured sessions and receive item-level review and actionable diagnostics.

ID	Goal	Success signal
G-02	Measure real learning progress, not only raw score.	Analytics distinguish mastery, independence, speed, recency, consistency and confidence.
G-03	Create a sustainable subscription product at Rp40,000–Rp50,000/month.	Trial-to-paid conversion, renewal and content engagement can be monitored.
G-04	Enable continuous content operations.	Authorized admins can create, review, publish, version and archive content without engineering deployment.
G-05	Operate securely and reliably for public internet access.	Security, privacy, backup, observability and incident controls satisfy defined acceptance criteria.

1.4 Non-Goals for Initial Release

- Live tutoring marketplace or teacher scheduling.
- AI-generated questions, AI grading or conversational tutor.
- Native iOS/Android applications.
- Institutional multi-tenant school administration.
- Official JLPT registration, certification or claims of affiliation.
- Offline-first synchronized learning.

1.5 Product Principles

- Evidence before vanity metrics.
- Explain every result.
- Do not punish returning learners by erasing history.
- Mobile-first accessibility and low-friction navigation.
- Content quality and auditability are product features.
- Security and privacy by design.

2. User and Access Model

2.1 Access States and Roles

Actor	Authentication	Authorization	Primary capabilities
Anonymous visitor	No	Public routes only	View landing/public pages, browse catalog previews, attempt one designated sample assessment and one sample learning material; results are ephemeral.
Registered trial user	Yes	USER + TRIAL entitlement	Same learning access as paid user for 24 hours after verified activation, subject to abuse controls.
Paid user	Yes	USER + ACTIVE subscription	Full question bank, exams, quizzes, notes, script drills, dashboards, history, invoices and settings.

Actor	Authentication	Authorization	Primary capabilities
Expired user	Yes	USER + EXPIRED entitlement	Can access account, billing, invoices and limited historical summaries; cannot start premium activities.
Administrator	Yes	ADMIN	Admin dashboard, content, media, users, payments, reports, CMS, SEO, audit logs and configuration according to granular permissions.

Role correction — "General user" should not be stored as a database role. It is an unauthenticated state. The authorization model therefore has two primary roles (USER and ADMIN), while entitlement states control trial, active, grace, expired, suspended and complimentary access.

2.2 Recommended Granular Admin Permissions

Permission group	Examples
Content	question.create, question.review, question.publish, material.manage, taxonomy.manage
Media	media.upload, media.delete, media.license.manage
Users	user.view, user.suspend, user.entitlement.grant
Finance	payment.view, refund.request, invoice.view, revenue.export
CMS/Marketing	page.manage, seo.manage, campaign.manage
Operations	audit.view, feature_flag.manage, system_setting.manage
Super admin	admin.manage, permission.assign, destructive operations approval

2.3 Core Personas

Persona	Needs	Risks/constraints
Beginner learner	Structured N5 foundations, hiragana/katakana drills, simple explanations, visible small wins.	May be overwhelmed by dense terminology and kanji.
Exam candidate	Timed simulations, competency breakdown, readiness trend, realistic review.	Needs score credibility and strict exam mode.
Returning learner	Rapid diagnosis of retained and decayed skills, comparison with previous periods.	Inactivity may make old scores misleading.
International learner	English-capable UI/content and understandable pricing.	Midtrans settles in IDR; local methods may be unavailable.
Content editor	Efficient authoring, validation, preview, versioning and media reuse.	Publishing mistakes can invalidate analytics.
Operations admin	User/payment visibility, audit evidence and reliable intervention tools.	Must not receive unnecessary access to sensitive data.

3. Scope and Information Architecture

3.1 Public Experience

- Landing page: value proposition, product capabilities, sample content, pricing, testimonials, FAQ and calls to action.
- Main/catalog page: searchable public catalog metadata with locked premium indicators.
- Public pages: About, Contact, FAQ, Terms, Privacy, Refund Policy, Cookie Notice, Accessibility statement and status/help pages.
- One admin-selected sample assessment and one sample material available without authentication; no durable personal analytics.

3.2 User Application Navigation

Navigation group	Pages/modules
Overview	Dashboard, readiness summary, current streak, recommendations, recent activity
Learn	Learning notes, collections, bookmarks, reading history
Practice	Quiz library, kana drills, kanji drills, custom practice
Exam	Exam simulations, scheduled mock exams, result history, review
Progress	Skill map, topic mastery, JLPT readiness, speed/accuracy, consistency, snapshots and comparisons
Account	Profile, security, notification preferences, billing, invoices and subscription

3.3 Admin Application Navigation

Navigation group	Pages/modules
Overview	Operational KPIs, content health, payment status, user activity, alerts
Content	Question bank, assessments, quizzes, materials, taxonomy, review queue, publishing calendar
Media	Images, audio, documents, metadata, usage references and licensing
Users	Accounts, entitlement, status, learning summary and support actions
Finance	Transactions, subscriptions, invoices, refunds, revenue reports, reconciliation
CMS & SEO	Pages, menus, hero, footer, FAQ, metadata, redirects, social sharing
Operations	Audit logs, jobs, webhooks, system health, feature flags and configuration

4. Learning Content Model

4.1 Taxonomy

Every question and material shall be mapped to a governed taxonomy. At minimum: JLPT level (N5–N1 or General), competency domain, topic, subtopic, grammar point, vocabulary set, script/kanji set, difficulty, content language, source provenance and status.

Domain	Illustrative subdomains
文字・語彙 (Characters/Vocabulary)	Hiragana, katakana, kanji recognition, reading, orthography, word formation, contextual vocabulary

Domain	Illustrative subdomains
文法 (Grammar)	Particles, conjugation, sentence patterns, ordering, usage in context
読解 (Reading)	Short notices, practical texts, medium passages, integrated comprehension
聴解 (Listening)	Task-based comprehension, key points, expressions, quick response; audio support can be staged
Pragmatics/Culture	Register, politeness, context-appropriate expression; marked as supplementary unless aligned to an exam blueprint

4.2 Question Types

Type	Initial release	Notes
Single-choice multiple choice	Required	One correct answer, distractors, explanation per option supported.
Multiple select	Optional/MVP+	Scoring rule must be explicit.
Ordering	Recommended	Useful for sentence construction.
Matching	Recommended	Vocabulary, reading and meaning relationships.
Text input	Limited	Kana/kanji normalization rules required.
Audio-based	Phase 2	Requires audio media, transcript governance and playback rules.
Image-based	Supported	Alt text and licensing metadata mandatory.

4.3 Content Lifecycle

1. Author creates draft and assigns taxonomy, source and learning objective.
2. System validates required fields, answer consistency, duplicate similarity and media references.
3. Reviewer evaluates linguistic accuracy, ambiguity, difficulty and explanation quality.
4. Publisher schedules or publishes a version. Published attempts bind to the immutable item version.
5. Corrections create a new version; prior versions remain available for historical result integrity.
6. Archived content cannot start new attempts but remains renderable for past results.

4.4 Learning Material Structure

- Title, slug, summary, target JLPT level and prerequisites.
- Learning objectives and estimated study time.
- Structured blocks: explanation, examples, vocabulary, grammar table, callout, image/audio and practice links.
- Related questions, materials and competency nodes.
- Version, author, reviewer, publication date, last review date and source references.
- Completion criteria: opened, engaged, completed, bookmarked and revisited. “Read” shall not be inferred from a page load alone.

5. Assessment and Quiz Requirements

5.1 Exam Preparation Mode

Requirement ID	Requirement
FR-EX-001	The system shall present a predefined or blueprint-generated set of questions under exam rules.
FR-EX-002	Answer correctness and explanations shall remain hidden until submission or expiry.
FR-EX-003	The system shall support timer, autosave, question navigation, flag-for-review and completion warning.
FR-EX-004	Submission shall produce score, accuracy, completion, time, topic distribution, independent performance and readiness interpretation.
FR-EX-005	Result review shall display the user answer, correct answer, explanation, option-level rationale where available, related material and taxonomy.
FR-EX-006	The system shall preserve the exact question versions and exam configuration used in the attempt.
FR-EX-007	Unexpected disconnect shall not lose acknowledged answers; server time is authoritative.
FR-EX-008	The system shall classify attempts as completed, expired, abandoned or invalidated.

5.2 Formative Quiz Mode

Requirement ID	Requirement
FR-QZ-001	A quiz may show immediate feedback after each answer.
FR-QZ-002	The learner may reveal an answer or explanation before committing only when the quiz configuration permits it.
FR-QZ-003	The system shall record reveal-before-answer, hint usage, answer changes, elapsed time and retry count.
FR-QZ-004	Final results shall separately report independent correct, assisted correct, independent incorrect, assisted incorrect, skipped and revealed-only items.
FR-QZ-005	Assisted answers shall not contribute equally to mastery as independent answers.
FR-QZ-006	The system shall provide a next-step recommendation based on weak competencies and related materials.

5.3 Kana and Kanji Drill

- Hiragana and katakana recognition, production, sound-to-character and character-to-sound modes.
- Kanji categorized by JLPT level, readings, meanings and configured vocabulary examples.
- Configurable duration, item count, direction, difficulty and inclusion filters.
- Spaced review queues may be introduced after the base event model is stable.
- Drill analytics contribute to script/kanji competency but remain distinguishable from full exam evidence.

5.4 Attempt State Machine

State	Entry	Exit
CREATED	Session initialized	IN_PROGRESS or CANCELLED
IN_PROGRESS	First item displayed	SUBMITTED, EXPIRED, ABANDONED
SUBMITTED	User submits all/partial answers	SCORING
EXPIRED	Server deadline reached	SCORING
SCORING	Server evaluates immutable item versions	COMPLETED or FAILED
COMPLETED	Result stored	REVIEWED
REVIEWED	Learner opens result review	Remains reviewed
INVALIDATED	Admin/system invalidates corrupt attempt	Excluded from mastery, retained for audit

6. Learning Analytics and Progress Intelligence

6.1 Analytics Principles

- Raw events are immutable; derived metrics are recalculable.
- Mastery is competency-specific, not a single global percentage.
- Recent independent evidence weighs more than old or assisted evidence.
- Speed is interpreted only with accuracy and item difficulty.
- Insufficient evidence must be shown as “not enough data”, not as low ability.
- Historical snapshots remain preserved for comparison.

6.2 Events to Capture

Event category	Examples
Assessment	attempt_started, item_viewed, answer_saved, answer_changed, item_flagged, attempt_submitted, attempt_expired
Assistance	hint_opened, answer_revealed, explanation_opened_before_commit, material_opened_from_item
Material	material_opened, active_read_seconds, section_completed, bookmark_added, material_completed, revisit
Drill	prompt_shown, response_submitted, response_correct, response_latency, session_completed
Account/subscription	trial_started, checkout_started, payment_pending, payment_settled, subscription_activated, expired
System integrity	client_disconnect, resume, duplicate_submit, scoring_version, invalidation

6.3 Core Metrics

Metric	Definition / interpretation
Raw accuracy	Correct answers ÷ scored answers.
Independent accuracy	Correct answers without prohibited reveal/hint ÷ independently attempted items.
Assisted accuracy	Correct answers where assistance occurred ÷ assisted attempts.
Coverage	Unique competency nodes with valid evidence ÷ targeted nodes.

Metric	Definition / interpretation
Mastery score	Recency- and difficulty-weighted evidence estimate per competency, bounded 0–100.
Confidence	Evidence sufficiency based on item count, diversity, recency and consistency.
Fluency	Accuracy-adjusted response speed against expected time bands.
Consistency	Stability across recent sessions; avoid conflating high variance with mastery.
Learning engagement	Meaningful active learning days, completed sessions, materials completed and revisits.
Readiness index	Blueprint-weighted competency mastery with minimum evidence gates; explicitly labeled as an estimate, not an official JLPT score.
Knowledge decay indicator	Expected retention risk based on time since valid evidence and prior stability.
Assistance dependency	Assisted items ÷ all attempted items, segmented by topic.

6.4 Recommended Mastery Calculation (Initial)

For each scored item, calculate an evidence value. Independent correct = +1.0; assisted correct = +0.35; independent incorrect = −0.55; assisted incorrect = −0.25; skipped/revealed-only = 0 with a dependency flag. Multiply by item difficulty factor (0.8–1.2), recency factor exp(−days/half-life), and quality factor based on response anomalies. Aggregate using a bounded logistic transformation. This is a product heuristic for MVP and must be calibrated with real data; it is not an Item Response Theory model.

Professional recommendation — After sufficient response volume and content calibration, migrate readiness estimation toward psychometric methods such as IRT/Rasch or Bayesian knowledge tracing. Do not market heuristic readiness as an official exam probability without validation.

6.5 Inactivity and One-Year Rule

The requested “reset after one year” should be implemented as an active-profile reset, not deletion. When no valid learning evidence exists for 365 days, the dashboard shall mark the current proficiency state as stale, set current confidence to “insufficient current evidence”, preserve previous snapshots, and invite a diagnostic session. New evidence initializes a new comparison period.

Data object	Retention behavior
Raw attempts and responses	Retain according to privacy policy and account lifecycle; never reset merely for inactivity.
Historical snapshots	Retain for user comparison while account exists.
Current active mastery profile	Mark stale after configured inactivity threshold; recalculate when new evidence arrives.
Public sample attempt	Ephemeral or short-lived anonymous aggregate; no personal profile.
Deleted account	Apply legal/privacy deletion workflow, with financial records retained only where legally required.

6.6 Dashboard Visualizations

- Overall readiness card with confidence and data freshness.
- JLPT/domain competency heatmap.
- Independent versus assisted accuracy.
- Accuracy and time trend by week/month.
- Weakest topics with recommended learning notes and quizzes.
- Strengths supported by sufficient recent evidence.
- Study consistency calendar and active days.
- Current period versus previous historical snapshot.
- Material-to-performance correlation presented as descriptive, not causal.

7. Subscription, Trial and Payment

7.1 Commercial Model

Item	Baseline
Monthly price	Product decision within Rp40,000–Rp50,000; recommended launch test: Rp49,000 including applicable taxes/fees presentation.
Trial	24 hours from successful email verification/first activation; one trial per verified identity and abuse-controlled device/payment signals.
Entitlement	Full premium learning features during trial and active paid period.
Grace	Optional 1–3 day grace for payment reconciliation; no new premium session after definitive expiry.
Renewal	Manual renewal for QRIS/VA/e-wallet unless a supported recurring/tokenized payment flow is explicitly implemented.
Refund	Policy-controlled and payment-method constrained; all actions audited.
Complimentary access	Admin-granted entitlement with reason and expiration.

7.2 Payment Methods

- QRIS.
- Supported e-wallet channels enabled in the Midtrans merchant account.
- Supported bank virtual accounts.
- Online-transaction-capable debit cards through supported card networks; operationally this is Midtrans card payment.
- Do not enable unrelated payment methods in Snap configuration for phase one.

7.3 Currency Handling

Midtrans requests shall be denominated in IDR. The product may detect locale and display an indicative converted price using a timestamped exchange-rate source, but shall state that the final charge is IDR and the issuing institution may apply its own conversion. True local-currency settlement requires an additional payment provider and is outside phase-one scope.

7.4 Payment State Machine

State	Meaning	Entitlement effect
CREATED	Checkout/order created	None
PENDING	Awaiting payer action or settlement	None or grace only
PAID/SETTLEMENT	Verified server-to-server notification and reconciliation	Activate/extend subscription idempotently
EXPIRED	Payment window expired	No activation
DENIED/CANCELLED	Provider/user rejected	No activation
REFUND_PENDING	Refund requested	Policy-dependent
REFUNDED/PARTIAL_REFUND	Refund confirmed	Adjust entitlement and ledger per policy
CHARGEBACK	Dispute/chargeback	Suspend or review according to risk policy

7.5 Payment Security Requirements

- Never trust browser payment success; verify Midtrans server notification signature and/or transaction status.
- Webhook processing shall be idempotent and replay-safe.
- Store provider references and masked metadata only; never store full card data.
- Maintain an append-only payment event log and internal ledger.
- Invoice numbers shall be unique and generated server-side.
- Reconciliation job shall identify pending inconsistencies and alert operations.

8. Functional Requirements — Account and User Experience

ID	Requirement	Priority
FR-AUTH-001	The system shall support email/password registration with email verification.	Must
FR-AUTH-002	The system shall support Google sign-in through the selected primary authentication framework.	Must
FR-AUTH-003	The system shall support password reset, session revocation and secure logout.	Must
FR-AUTH-004	The system shall rate-limit registration, login, verification and reset endpoints.	Must
FR-AUTH-005	Admin accounts shall require multi-factor authentication before production access.	Must
FR-USER-001	Users shall edit profile, preferred interface language, timezone and study target.	Must
FR-USER-002	Users shall view active sessions and revoke other sessions.	Should
FR-USER-003	Users shall export a portable copy of their learning data.	Should
FR-USER-004	Users shall request account deletion subject to retention disclosures.	Must
FR-DASH-001	Dashboard shall show current mastery, freshness, confidence, recent activity and recommended next actions.	Must
FR-DASH-002	Dashboard shall clearly distinguish insufficient evidence from poor performance.	Must

ID	Requirement	Priority
FR-NOTIF-001	Users shall manage email notification preferences for learning reminders, billing and security.	Should

9. Functional Requirements — Content and Administration

ID	Requirement	Priority
FR-CNT-001	Admins with permission shall create, edit, duplicate, preview, review, publish, schedule, unpublish and archive questions.	Must
FR-CNT-002	Published questions shall be immutable by version; edits create a new version.	Must
FR-CNT-003	The authoring form shall validate that answer keys, explanations and options are internally consistent.	Must
FR-CNT-004	Questions shall support taxonomy, difficulty, estimated time, source, language and media metadata.	Must
FR-CNT-005	Admins shall build assessments using fixed items or governed blueprint rules.	Must
FR-CNT-006	Admins shall preview an assessment exactly as a learner.	Must
FR-MAT-001	Admins shall create modular learning materials using safe structured blocks or sanitized rich text.	Must
FR-MED-001	Media management shall track file type, size, hash, alt text, license, owner, references and lifecycle state.	Must
FR-CMS-001	Authorized admins shall edit public page content, navigation, FAQ, metadata and social preview fields.	Must
FR-SEO-001	Public pages shall support canonical URL, title, description, Open Graph, sitemap inclusion and robots directives.	Should
FR-ADM-001	All sensitive admin actions shall create audit records containing actor, action, target, timestamp, reason and result.	Must
FR-ADM-002	Destructive or high-risk actions shall require confirmation and, where configured, dual approval.	Should

10. System Requirements Specification (SRS)

10.1 System Context

Onigiroku consists of a public web application, authenticated learner application, admin application, server-side application layer, relational database, Redis, object storage, background job workers, email provider and Midtrans integration. The browser never directly receives database, storage or payment secrets.

10.2 Recommended Architecture

Layer	Technology/direction
Web application	Next.js with TypeScript; server rendering for public SEO pages; authenticated app routes with server-side authorization.
UI	Tailwind CSS, shadcn/ui, Lucide React, Sonner notifications, Framer Motion used selectively with reduced-motion support.

Layer	Technology/direction
Validation/state	Zod at every trust boundary; TanStack Query for server state; Zustand only for limited ephemeral client state.
Application structure	Modular monolith organized by auth, content, assessment, analytics, billing, CMS, media and administration domains.
Database	PostgreSQL accessed through Drizzle ORM; migration workflow mandatory; pgAdmin is an operations tool, never public.
Cache/jobs	Redis for rate limiting, short-lived cache, distributed locks and queues; not the source of truth.
Object storage	SeaweedFS exposed through private S3-compatible endpoint; presigned upload/download; malware scanning pipeline.
Authentication	Better Auth as primary system with email/password and Google provider. Firebase Authentication removed from baseline to avoid split identity/session ownership.
Email	Nodemailer transport abstraction with production transactional email provider; SPF, DKIM and DMARC configured.
Payments	Midtrans Snap or Core API selected based on UX/control; server-side token creation and verified webhook handling.
Observability	Structured logs, metrics, traces/error monitoring, uptime checks and alerting.
Deployment	Containerized workloads behind TLS reverse proxy/WAF; separate development, staging and production environments.

10.3 Why Not Better Auth + Firebase Auth Together?

Both products can own identities, sessions and Google sign-in. Running them simultaneously creates duplicate user identifiers, account-linking complexity, inconsistent revocation, two security configurations and harder support. Better Auth already supports email/password and Google OAuth. Firebase may remain available for unrelated future services, but it should not be a second authentication authority without a documented architectural reason.

10.4 Logical Components

Component	Responsibilities
Identity & Access	Registration, verification, login, sessions, OAuth, MFA, roles and permissions.
Content Service	Taxonomy, questions, versions, materials, media references and publishing.
Assessment Engine	Session creation, item delivery, autosave, timing, scoring and immutable results.
Learning Event Collector	Validates and persists event stream with idempotency.
Analytics Processor	Updates competency evidence, snapshots, recommendations and dashboard projections.
Billing Service	Orders, payment webhooks, ledger, subscription entitlements, invoices and reconciliation.
CMS Service	Public pages, navigation, FAQ and SEO fields.
Notification Service	Transactional email and queued learning reminders.
Admin Operations	User actions, reporting, audit logs, jobs and feature flags.

10.5 Key Data Entities

Domain	Entities
Identity	users, accounts, sessions, verification_tokens, roles, permissions, user_roles, admin_mfa
Subscription	plans, prices, trials, subscriptions, entitlements, orders, payments, payment_events, invoices, refunds
Content	taxonomies, competencies, questions, question_versions, options, explanations, materials, material_versions, media_assets, content_reviews
Assessment	assessments, assessment_versions, assessment_items, attempts, attempt_items, responses, assistance_events, score_results
Analytics	learning_events, competency_evidence, mastery_profiles, mastery_snapshots, recommendations, study_daily_facts
CMS	pages, page_versions, menus, faqs, seo_metadata, redirects
Operations	audit_logs, webhook_receipts, job_runs, feature_flags, system_settings, notifications

10.6 Data Integrity Rules

- Every attempt references immutable assessment and question versions.
- A response write includes attempt ID, item ID, client sequence and idempotency key.
- Scoring executes server-side with a versioned scoring algorithm.
- Payment order IDs are unique; webhook event processing uses unique provider event/transaction keys.
- Entitlement changes are append-only events projected into current status.
- Admin edits never rewrite historical audit records.
- Timestamps are stored in UTC and rendered in user timezone.

10.7 API Requirements

Area	Representative endpoints/operations
Auth	Framework-managed auth routes; session and permission middleware.
Catalog	GET /api/catalog, /materials, /assessments with filters and pagination.
Attempts	POST attempt; GET next/current item; PUT response; POST submit; GET result/review.
Events	POST batched learning events with schema version and idempotency.
Analytics	GET dashboard, competency map, history, comparison and recommendations.
Billing	POST checkout; GET subscription/invoices; POST Midtrans webhook; admin reconciliation.
Admin content	CRUD draft resources, review, publish, schedule, archive and version diff.
Media	Create presigned upload, finalize asset, scan status and reference usage.
CMS	Public read endpoints with cache; protected admin mutation endpoints.

10.8 API Standards

- JSON over HTTPS; consistent envelope and machine-readable error codes.
- OpenAPI specification maintained with implementation.
- Cursor pagination for large collections.
- Idempotency keys for create-attempt, submission, payment and webhook operations.
- Optimistic concurrency/version checks for admin edits.
- No sensitive implementation details in errors.

11. Non-Functional Requirements

Category	Requirement / target
Availability	Production monthly availability target ≥99.5% for MVP, excluding announced maintenance; evolve to ≥99.9% after stabilization.
Performance	Public LCP p75 ≤2.5s on representative mobile network; authenticated API p95 ≤500ms for ordinary reads and ≤1s for ordinary writes, excluding third-party latency.
Autosave	Acknowledged answer persistence within 1 second under normal conditions; retry safely after network loss.
Scalability	Stateless web replicas; background workers horizontally scalable; database indexes and query budgets reviewed.
Accessibility	WCAG 2.2 AA target: keyboard operation, focus visibility, contrast, labels, alt text, reduced motion and semantic structure.
Responsive design	Support modern mobile, tablet and desktop viewports; assessment interaction usable from 360px width.
Localization	Architecture supports Indonesian and English UI; Japanese content with proper fonts, ruby/furigana strategy and locale-aware formatting.
Reliability	No single browser event shall be the sole source of payment or result completion truth.
Backup	Encrypted automated PostgreSQL backups plus object-storage backup policy; restore tests conducted regularly.
Recovery	Initial target RPO ≤24h and RTO ≤4h; tighten for paid production after operational validation.
Maintainability	Typed code, linting, tests, migrations, code review and architecture decision records.
Privacy	Data minimization, consent/notice, retention schedule, export/deletion workflow and access logging.
Observability	Correlation IDs, structured logs, error tracking, metrics and alert thresholds without logging secrets or answer data unnecessarily.

12. Security and Privacy Requirements

12.1 Security Baseline

The application should use OWASP ASVS 5.0 as the verification baseline and consider the OWASP Top 10 current release during threat modeling. Security controls shall be tested in CI and staging, not treated as documentation only.

Control area	Mandatory controls
Authentication	Strong password policy, breached-password defense where practical, email verification, MFA for admins, session rotation, secure cookies, CSRF defense and session revocation.
Authorization	Deny-by-default server-side checks; RBAC plus entitlement checks; object-level authorization on every resource.
Input/output	Zod validation, parameterized ORM queries, contextual output encoding, sanitized rich text, strict file validation.
Browser security	TLS, HSTS, CSP, frame-ancestors, Referrer-Policy, Permissions-Policy and secure cookie attributes.
Abuse prevention	Rate limits, bot controls, trial abuse detection, credential-stuffing monitoring and account lock/risk response.
Secrets	Environment/secret manager, rotation, least privilege, no secrets in client bundles, source control or logs.
Files	Allowlist MIME and extension, maximum size, content sniffing, malware scan, quarantine, random object keys and private-by-default access.
Payments	Signature verification, status verification, idempotency, reconciliation, masked data and no storage of cardholder data.
Admin	Separate routes, MFA, short sessions, re-authentication for high-risk actions and complete audit trail.
Database	Private network, TLS where applicable, least-privileged app account, encrypted backups and restricted pgAdmin.
Supply chain	Lockfile, dependency scanning, secret scanning, image scanning, SBOM and controlled updates.
Incident response	Severity model, on-call contacts, containment steps, evidence preservation and user notification procedure.

12.2 Privacy Classification

Class	Examples	Handling
Public	Published materials and public pages	Cacheable and indexable as configured.
Internal	Content drafts, operational dashboards	Authenticated and permission-restricted.
Personal	Name, email, profile, learning history	Purpose-limited, access-controlled, exportable/deletable.
Financial	Payment references, invoices, billing status	Restricted, audited and retained per legal/accounting policy.
Security-sensitive	Session tokens, reset tokens, secrets, MFA data	Encrypted/hashed as appropriate; never exposed or logged.
Derived profiling	Mastery, weaknesses, readiness and recommendations	Explainable to user, access-controlled and covered by privacy notice.

12.3 Threat Scenarios to Test

- User accesses another user's attempt, invoice or analytics by changing an identifier.

- Expired/trial user calls premium APIs directly.
- Admin without publish permission publishes content.
- Webhook spoofing or replay activates subscription.
- Client manipulates timer, score or answer key.
- Stored XSS through learning material/CMS content.
- Malicious file uploaded through media manager.
- Trial farming through repeated accounts.
- Enumeration through login/reset responses.
- Sensitive Japanese answer keys leaked through prefetch, source maps, caches or client payloads.

13. UX, Visual Design and Brand System

13.1 Brand Interpretation

The supplied mark uses a vivid red and a dark charcoal outline around a white onigiri symbol. The interface should use the red as a controlled brand accent rather than a full-page dominant background, preserving readability and professional credibility.

13.2 Recommended Color Palette

Token	Hex	Usage	Accessibility note
Brand Primary	#E60012	Primary buttons, active states, key brand moments	Use white text only when contrast is verified; prefer darker hover #C90010.
Brand Vivid	#FF0000	Logo fidelity, small highlights, illustrations	Avoid large text blocks/background dominance.
Charcoal	#434040	Primary text, navigation, borders with opacity	Main neutral from logo.
Ink	#18181B	High-emphasis text	High contrast on light surfaces.
Rice White	#FFFFFF	Cards and core surfaces	Pair with subtle borders/shadows.
Warm Canvas	#FAF8F6	Application background	Creates softer Japanese-inspired visual tone.
Soft Gray	#F3F4F6	Secondary panels, disabled areas	Do not use for low-contrast body text.
Success	#15803D	Correct answers, paid status, positive change	Never rely on color alone; include icon/text.
Warning	#B45309	Stale progress, pending payment	Include descriptive label.
Error	#B91C1C	Errors, incorrect answer, destructive action	Separate from brand red by context and tone.
Info	#1D4ED8	Informational notices and links	Use consistent link treatment.

13.3 Typography

- Primary UI: Inter, Geist or equivalent highly legible sans serif.
- Japanese content: Noto Sans JP; optional Noto Serif JP for editorial headings only.
- Use font fallback that preserves kana/kanji glyph coverage.
- Body text minimum 16px on learning content; assessment options designed for touch.

13.4 UI Patterns

- Rounded cards and controls with moderate radius (10–14px), not excessively playful.
- Generous whitespace; clear content hierarchy; left navigation on desktop and bottom/tab navigation where justified on mobile.
- Progress colors paired with labels and icons.
- Result pages prioritize interpretation and next action over celebratory animation.
- Framer Motion used for subtle transitions; honor prefers-reduced-motion.
- Use furigana/ruby rendering where content requires it.

13.5 Key Screens

Screen	Required content
Landing	Hero, credibility, how it works, capabilities, sample preview, pricing, FAQ and CTA.
User dashboard	Readiness, confidence/freshness, recommended next action, streak/consistency, recent work and weak topics.
Assessment player	Timer, progress, item content, answer area, flag, navigation, autosave state and accessibility controls.
Result review	Score summary, independent/assisted breakdown, domain analysis, item review and related learning.
Material reader	Outline, objectives, content blocks, progress, bookmarks and linked practice.
Admin question editor	Metadata, item body, options/key, explanation, preview, validation, version and workflow.
Admin operations	Filterable tables, exports, safe actions, audit context and status indicators.

14. Reporting and Administration

14.1 Admin KPIs

- Registered, verified, trial, active paid, expired and suspended users.
- Trial starts, trial completion, conversion and conversion by cohort.
- Monthly recurring revenue proxy, gross collections, refunds and net collections.
- Payment success/pending/expiry by channel.
- Daily/weekly active learners, session completion and material engagement.
- Content coverage by JLPT/domain, stale review dates, flagged items and item-performance anomalies.
- Operational errors, failed jobs, webhook delays and storage usage.

14.2 Content Quality Analytics

Signal	Potential action
Very high/low correct rate	Review difficulty classification or ambiguity.
Distractor never selected	Improve distractor plausibility.
High answer-change rate	Review wording or option similarity.
Long response time with high accuracy	Potential reading load rather than conceptual difficulty.
High reveal dependency	Improve prerequisite material or explanation.
Performance shift after version change	Compare versions and investigate regression.
User reports	Queue for editorial review with severity and affected attempts.

14.3 Audit Log Requirements

Audit logs shall be append-only from the application perspective and include actor, role, source IP/device metadata where appropriate, action, target type/ID, before/after summary for permitted fields, reason, correlation ID and outcome. Sensitive values and secrets shall be redacted.

15. Testing and Quality Strategy

Test layer	Scope
Unit	Scoring, entitlement, mastery calculations, validation and utility logic.
Integration	Database repositories, Redis behavior, storage lifecycle, email adapter and Midtrans signature/status handling.
API contract	OpenAPI validation, error schema, permissions and idempotency.
Component	Assessment controls, forms, tables, charts and accessible interactions.
End-to-end	Registration, trial, checkout, entitlement, exam, quiz, result review, material completion and admin publishing.
Security	SAST, dependency/secret/image scans, authorization tests, XSS/CSRF, rate limits, upload and webhook abuse.
Performance	Load test attempt autosave, submission/scoring, dashboards and admin lists.
Accessibility	Automated checks plus keyboard and screen-reader manual flows.
Content QA	Linguistic review, answer validation, rendering of Japanese text/furigana and media transcript checks.
Disaster recovery	Database and object restore drill with documented results.

15.1 Release Gates

- No unresolved critical/high security defect.
- All Must requirements for release milestone accepted.
- Payment sandbox and production checklist passed.
- Backup restore demonstrated.
- Admin MFA and audit logging active.
- Privacy, terms, refund and billing disclosures published.
- Monitoring and rollback procedure validated.

16. DevOps and Operational Requirements

- Separate dev, staging and production accounts/configuration.
- Infrastructure and application configuration documented and reproducible.
- CI: typecheck, lint, unit/integration tests, build, migrations check, security scans.
- CD: immutable image, staging deployment, smoke test, controlled production promotion and rollback.
- Database migrations are forward-reviewed, backed up and observable.
- Blue/green or rolling deployment should preserve active assessment sessions.
- Scheduled jobs: analytics projection, payment reconciliation, content review reminders, backup verification and data retention.
- Feature flags permit staged rollout of analytics and billing behavior.

16.1 Monitoring Alerts

Alert	Example threshold
Availability	Repeated failed health checks or elevated 5xx rate.
Assessment durability	Autosave failure rate exceeds baseline.
Payments	Webhook failures, signature failures or pending reconciliation backlog.
Queue	Oldest job age or dead-letter count above threshold.
Database	Connection saturation, storage growth, replication/backup failure.
Object storage	Capacity, scan failures or inaccessible assets.
Security	Login abuse, admin anomalies or permission-denied spikes.

17. Delivery Roadmap

Phase	Duration estimate	Deliverables
0 — Discovery & foundation	2–3 weeks	Final taxonomy, content standards, wireframes, architecture decisions, environments, repository, CI and threat model.
1 — MVP core	8–12 weeks	Public site, Better Auth, trial/entitlement, content authoring, materials, quiz/exam player, result review, basic dashboard, Midtrans one-time payment, admin basics.
2 — Analytics & drills	6–8 weeks	Learning event pipeline, mastery model, history comparison, kana/kanji drills, recommendations, richer reports and operational dashboards.
3 — Production hardening	4–6 weeks	Security verification, performance, accessibility, backup/restore, reconciliation, audit enhancements, content QA and launch operations.
4 — Growth	Ongoing	Audio/listening, advanced psychometrics, spaced repetition, referrals, multilingual expansion and institutional options.

17.1 MVP Prioritization

Must	Should	Later
Auth, trial, subscription entitlement; question/material CMS; exam and quiz flows; immutable results; basic analytics; Midtrans; admin audit; responsive and secure deployment.	Google login, bookmarks, kana/kanji drill, recommendations, exports, scheduled publishing, richer SEO and operations reports.	Listening content, IRT/BKT, social/community, native apps, institutional tenancy, AI features and multiple payment processors.

18. Acceptance Criteria — Critical Journeys

Journey	Acceptance criteria
Registration and trial	Verified new user receives exactly one 24-hour trial; duplicate callbacks/actions do not extend it; unverified user cannot consume premium entitlement.
Paid checkout	A successful verified Midtrans settlement activates or extends entitlement once; browser-only success cannot activate; duplicate webhooks are harmless.
Exam attempt	Answers survive refresh/reconnect; server deadline is enforced; answer keys are not delivered before completion; result binds to immutable versions.
Quiz assistance	Reveal/hint events are recorded; final report accurately separates independent and assisted outcomes.
Progress dashboard	Metrics reconcile to source events; stale/insufficient states are explicit; historical snapshot remains visible after one-year inactivity.
Content publishing	Only permitted admin publishes; required validation passes; historical attempts continue rendering the old version.
Authorization	A user cannot access another user's attempt, invoice, profile or analytics through direct API manipulation.
File upload	Invalid, oversized or malicious file is rejected/quarantined; unreferenced private file is not publicly enumerable.
Account deletion	User request follows confirmation and retention policy; access is revoked; deletion/anonymization actions are audited.

19. Risks and Mitigations

Risk	Impact	Mitigation
Content accuracy/ambiguity	Loss of trust and invalid analytics	Editorial workflow, versioning, reviewer accountability, item analytics and user reporting.
Dual authentication architecture	Identity inconsistency and security defects	Use one authentication authority; Better Auth baseline.
Misleading readiness score	Users overestimate exam capability	Confidence gates, transparent methodology, calibration and no official-score claims.

Risk	Impact	Mitigation
Trial abuse	Revenue leakage	Verification, rate limits, device/risk signals and one-trial policy.
Payment mismatch	Access/support incidents	Verified webhook, idempotency, ledger and reconciliation.
Client-side cheating	Invalid results	Server scoring/time authority, no preloaded keys and anomaly events.
Scope expansion	Delayed launch	Strict MVP gates and roadmap separation.
Self-hosted storage/ops complexity	Availability and data loss	Private networking, monitoring, backups, restore drills and documented operations.
Foreign-currency expectation	Checkout confusion	Display estimate with timestamp; charge IDR; add international processor only as separate scope.
One-year reset interpretation	Loss of user trust	Preserve history; reset only current evidence confidence/profile.

20. Open Product Decisions

Decision	Options / recommendation
Launch price	Rp40k, Rp45k or Rp49k. Recommend Rp49k list price with launch promotion only if unit economics support it.
Trial start trigger	Email verification vs first premium activity. Recommend verification plus first activation, with an expiry cap to prevent dormant trials.
Expired-user history access	Full read-only history vs summary only. Recommend read-only history, while premium review details may be gated according to policy.
Exam content blueprint	Manual fixed forms vs dynamic blueprint. Begin fixed/versioned forms, then add governed blueprint generation.
Listening scope	Launch vs phase 2. Recommend phase 2 unless sufficient licensed audio and QA capacity already exist.
Interface language	Indonesian only vs Indonesian + English. Architect bilingual from start; launch Indonesian with essential English public/payment copy if capacity is limited.
Data retention	Exact legal and operational durations require counsel/accounting review before launch.
Refund policy	Define eligibility window, consumed-service rules and payment-method constraints before production.

21. Reference and Standards Notes

The following official sources informed architectural constraints and security/payment recommendations. They should be rechecked during implementation because external products and standards evolve.

7. Midtrans Documentation — Payment methods, Snap/Core API, QRIS, virtual accounts, e-wallet and card payments: https://docs.midtrans.com/
8. Midtrans — Foreign currency guidance: charge requests must be sent in IDR; conversion may be displayed at checkout: https://docs.midtrans.com/docs/can-i-receive-payments-using-other-currency-than-idr
9. Midtrans — Subscription API currently identifies IDR as the supported currency: https://docs.midtrans.com/reference/update-subscription
10. Midtrans — Online-capable debit cards are processed through supported card networks: https://docs.midtrans.com/docs/coreapi-card-payment-integration
11. Better Auth — Email/password and social provider support, including Google: https://www.better-auth.com/docs/
12. Firebase Authentication — Google sign-in documentation (reviewed to assess overlap): https://firebase.google.com/docs/auth/web/google-signin
13. OWASP Application Security Verification Standard (ASVS) 5.0: https://owasp.org/www-project-application-security-verification-standard/
14. OWASP Top Ten Web Application Security Risks: https://owasp.org/www-project-top-ten/

Appendix A — Suggested Repository Structure

A domain-oriented modular monolith keeps the initial system manageable while retaining clean boundaries.

```
apps/
  web/                 # Next.js public, learner and admin UI
packages/
  auth/                # Better Auth configuration and authorization
  db/                  # Drizzle schema, migrations and repositories
  content/             # Taxonomy, questions, materials, publishing
  assessment/          # Attempts, timing, scoring and review
  analytics/           # Events, mastery, snapshots, recommendations
  billing/             # Midtrans, ledger, entitlement and invoices
  media/               # SeaweedFS/S3 adapter and scanning workflow
  cms/                 # Public content and SEO
  notifications/       # Email templates and queue handlers
  ui/                  # Shared design system
  observability/       # Logging, tracing and metrics
workers/
  jobs/                # Analytics, email, reconciliation, media scan
infra/                 # Containers, deployment and operational docs
tests/                 # E2E, fixtures and security tests
```

Appendix B — Definition of Done

- Requirement and acceptance criteria satisfied.
- Authorization and entitlement tests included.
- Input validation and error paths covered.
- Observability added without sensitive logging.
- Database migration reviewed and rollback/forward strategy documented.
- Responsive and keyboard behavior verified.
- Japanese glyphs and content rendering verified.

- Documentation/OpenAPI updated.
- Security scans pass and code review completed.
- Staging smoke test succeeds.

Appendix C — Glossary

Term	Definition
Assisted attempt	An item response influenced by an answer reveal, hint or other configured assistance.
Competency node	A measurable Japanese-language skill/topic in the product taxonomy.
Entitlement	The current authorization to consume premium product capabilities.
Immutable version	A published content snapshot that is never overwritten and remains tied to historical attempts.
Mastery	A derived estimate of capability for a specific competency based on weighted evidence.
Readiness index	An internal, non-official estimate combining blueprint-weighted mastery and evidence confidence.
Recency decay	Reduction in the influence of older evidence when estimating current ability.
Snapshot	A preserved derived profile for historical comparison.
Trial	Time-limited premium entitlement for a verified new user.



End of Document